

CSE331: Automata and Computability

Assignment 2

Deadline: 12/08/2026 11:59 PM

Show the following language is not regular using pumping lemma

1. $L = \{ 0^{2n}1^n \mid n \geq 0 \}$, where $\Sigma = \{0, 1\}$.

Check quiz 3 solve

2. $L = \{ w \mid w \text{ is not a palindrome} \}$, where $\Sigma = \{0, 1\}$.

Solⁿ:

If a language is not regular, its complement is also not regular. So we will first prove that $\bar{L}$ is not regular.

$\bar{L} = \{ w \mid w \text{ is a palindrome} \}$, where $\Sigma = \{0, 1\}$

- Suppose $\bar{L}$ is regular. Then it must satisfy the pumping property.

- $w = 0^s 1 0^s$ where $|w| \geq s$
 $2s+1 \geq s$

- $w = xyz$ where $|xy| \leq s$ and $|y| \geq 1$

$\begin{array}{ccc} \swarrow & \downarrow & \downarrow \\ 0^p & 0^q & 0^{s-p-q} 1 0^s \end{array}$

- Also $xy^p z$ must belong to L for all $p \geq 0$.

$$\begin{aligned} p=2, xy^2 z &= 0^p 0^q 0^q 0^{s-p-q} 1 0^s \\ &= 0^p 0^q 0^q 0^s 0^q 0^q 1 0^s \\ &= 0^q 0^s 1 0^s \\ &= 0^{q+s} 1 0^s \notin L \end{aligned}$$

- Contradiction! Hence L is not regular.

CSE331: Automata and Computability

Assignment 2

Deadline: 12/08/2026 11:59 PM

3. $L = \{a^m b^n \mid m \neq n\}$, where $\Sigma = \{a, b\}$.

- Assuming : that L is regular and satisfies the pumping property

- Choosing w : $w = a^s b^{s+1}$ [$s + s + 1 \geq s$]

- Splitting : $w = xyz$ [$|xy| \leq s$ and $|y| \geq 1$]

$$x = a^p, \quad y = a^q, \quad z = a^{s-p-q} b^{s+1}$$

- Checking if $xy^i z \in L$ for all $i \geq 0$:

- $i=1$: $xyz = \cancel{a^p} \cancel{a^q} a^s \cancel{a^{-p-q}} b^{s+1} = a^s b^{s+1}$

- $i=2$: $xyyz = \cancel{a^p} a^q \cancel{a^q} a^s \cancel{a^{-p-q}} b^{s+1} = a^{s+q} b^{s+1}$

- $i = \frac{s+1}{q} + 1$: $xy^{\left(\frac{s+1}{q} + 1\right)} z = a^p (a^q)^{\left(\frac{s+1}{q} + 1\right)} a^{s-p-q} b^{s+1}$
 $= a^p a^{(s+1+q)} a^s a^{-p-q} b^{s+1}$
 $= \cancel{a^p} a^{s+1} \cancel{a^q} a^s \cancel{a^{-p-q}} b^{s+1}$
 $= a^{s+1} b^{s+1} \notin L$

Contradiction, when $i = \frac{s+1}{q} + 1$, w has equal number of a 's and b 's.

L is not a regular language.

CSE331: Automata and Computability

Assignment 2

Deadline: 12/08/2026 11:59 PM

4. $L(M) \rightarrow \{0^i 1^j 2^k \mid i, j, k \geq 0 \text{ and if } i = 1 \text{ then } j = k, \text{ otherwise } j \neq k\}$

Solution. Assume for the sake of contradiction that L is regular. Since it's regular, it has a pumping length, say p . Now we pick the string

$$w = 01^p 2^p \in L.$$

Then $|w| = 2p + 1 \geq p$. So w can be split into $w = xyz$ such that

- (i) $|y| > 0$,
- (ii) $|xy| \leq p$, and
- (iii) $xy^i z \in L$ for every $i \geq 0$.

We have **three cases**:

1. $x = \epsilon, y = 0$.

In this case,

$$xy^2 z = 001^p 2^p = 0^2 1^p 2^p \notin L,$$

because here $i \neq 1$, but $j = k$.

2. $x = \epsilon, y = 011 \dots 1$.

In this case,

$$xy^2 z = 011 \dots 1 \ 011 \dots 1 \ 11 \dots 122 \dots 2 \notin L,$$

because there is a 0 after 1s.

3. $x \neq \epsilon$.

In this case, x contains a 0. Since $|xy| \leq p$, and after the first 0, the next p symbols are all 1s, then y consists of 1s only.

$$xy^2 z = 01^{p+|y|} 2^p \notin L,$$

since $p + |y| \neq p$.

Thus we get a contradiction in all cases. Therefore, L can't be regular. ■

Context free grammar

1. $L = \{w \in \{0,1\}^* : w = 0^n 1^m, n, m \geq 0 \text{ and } 2n = 3m\}$

$$S \rightarrow 000S11 \mid \epsilon$$

2. $L = \{w \in \{0,1\}^* : \text{the length of } w \text{ is divisible by } 3\}$

$$S \rightarrow psp \mid \epsilon$$

$$p \rightarrow 0 \mid 1$$

CSE331: Automata and Computability

Assignment 2

Deadline: 12/08/2026 11:59 PM

3. $L = \{ w \in \{0,1\}^* : w_1 \# w_2, \text{ the number of 0's in } w_1 \text{ is equal to number of 1's in } w_2 \}$

$$S \rightarrow 1S \mid S0 \mid 0S1 \mid \#$$

4. $L = \{ w \in \{0,1,2\}^* : w = 0^i 2^j 1^k, \text{ where } i = k, i, k \geq 1 \text{ and } j \geq 2 \}$

$$\begin{aligned} w &= 0^i 2^j 1^k \\ &= 0^k 2^j 1^k \end{aligned} \quad \begin{aligned} i, k &\geq 1 \\ j &\geq 2 \end{aligned}$$

$$\begin{aligned} S &\rightarrow 0P1 \\ P &\rightarrow 0P1 \mid y \\ y &\rightarrow 22x \\ x &\rightarrow 2x \mid \epsilon \end{aligned}$$

5. $L = \{ w \in \{0,1,2\}^* : w = 0^i 2^j 1^k, \text{ where } i \text{ is multiple of two, } k \text{ is two more than multiple of 3, } j = k+i \text{ and } i, j, k \geq 0 \}$

$$\begin{aligned} w &= 0^i 2^j 1^k \\ &= 0^{2i} 2^{3k+2+2i} 1^{3k+2} \\ &= 0^{2i} 2^i 2^k 1^k 1^2 \end{aligned} \quad \begin{aligned} j &= k+i \\ &= 3k+2+2i \\ &= 2i+3k+2 \end{aligned}$$

$$\begin{aligned} S &\rightarrow xy \\ x &\rightarrow 00x22 \mid \epsilon \\ y &\rightarrow 222y111 \mid 11 \end{aligned}$$

6. $L = \{ w \in \{0,1\}^* : w \text{ is not a palindrome} \}$

$$S \rightarrow 0S0 \mid 1S1 \mid 0A1 \mid 1A0$$

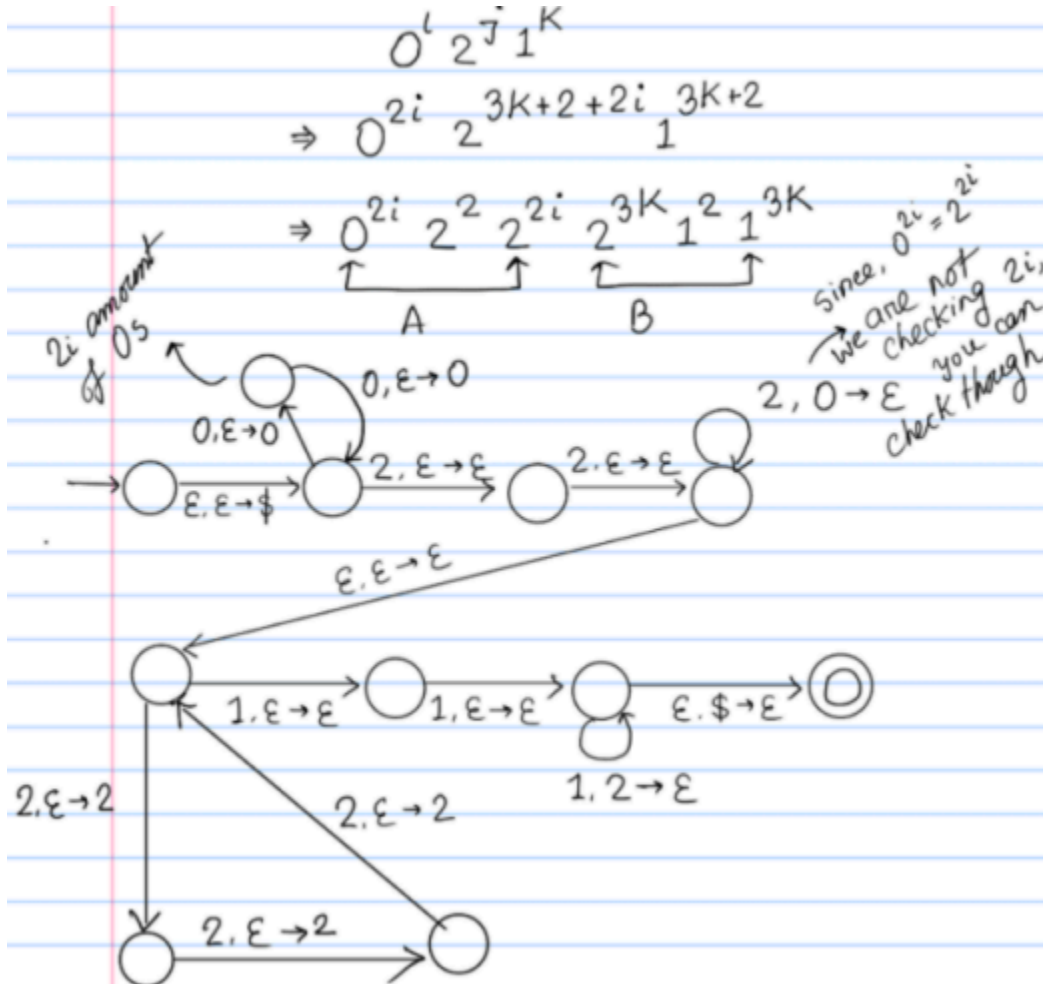
$$A \rightarrow 0A \mid 1A \mid \epsilon$$

CSE331: Automata and Computability

Assignment 2

Deadline: 12/08/2026 11:59 PM

2. $L = \{ w \in \{0,1,2\}^* : w = 0^i 2^j 1^k, \text{ where } i \text{ is multiple of two, } k \text{ is two more than multiple of 3, } j = k+i \text{ and } i,j,k \geq 0 \}$

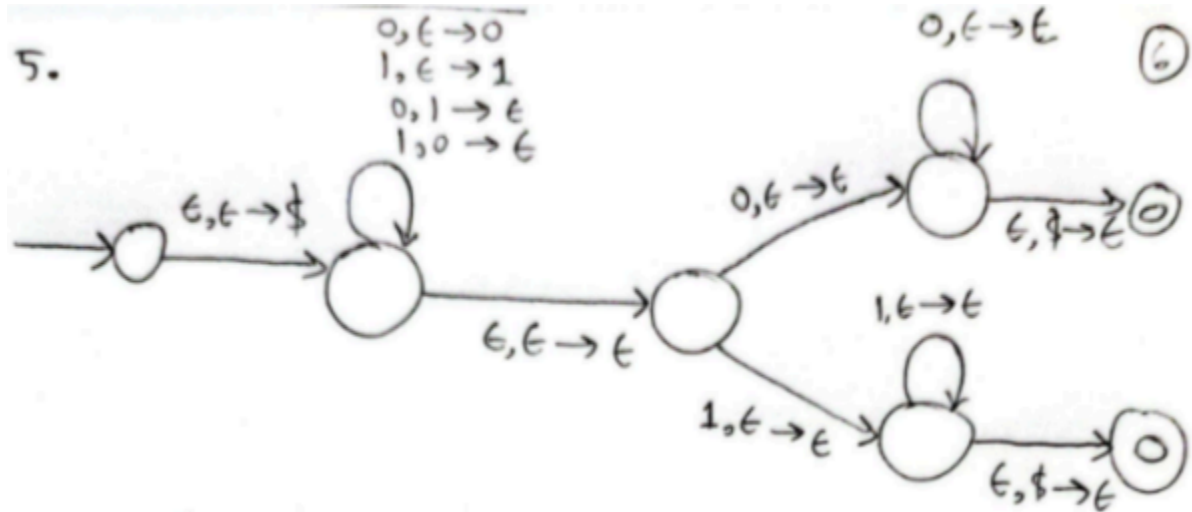


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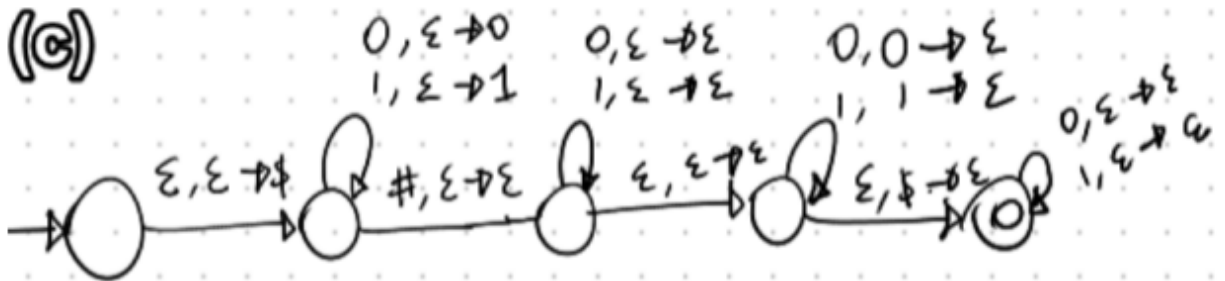
Assignment 2

Deadline: 12/08/2026 11:59 PM

3. $L = \{ w \in \{0,1\}^* : \text{the number of 0s and 1s are different in } w \}$



4. $L = \{ w\#x : w, x \in \{0,1\}^* \text{ and } w^R \text{ is a substring of } x \}$



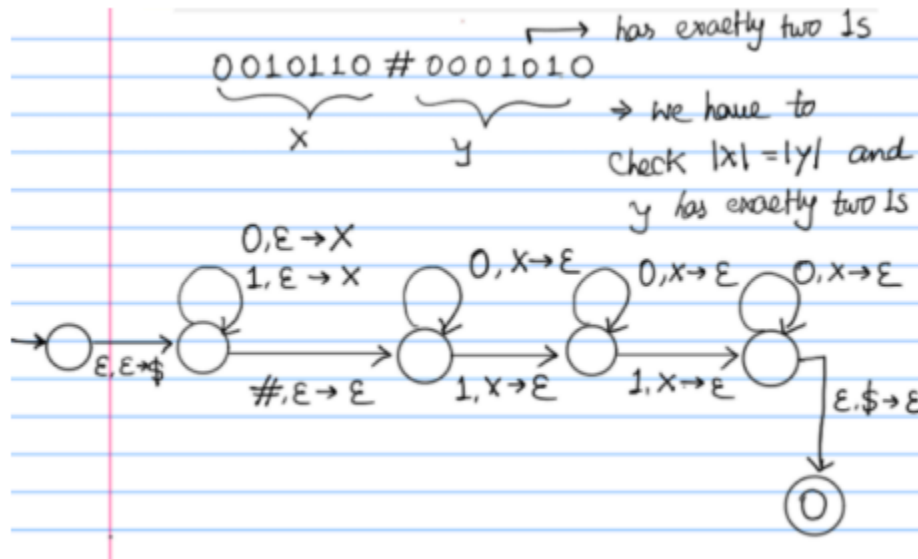
5. Recall that for a string w , $|w|$ denotes the length of w . $\Sigma = \{0,1\}$
 $L1 = \{ w \in \Sigma^* : w \text{ contains exactly two 1s} \}$
 $L2 = \{ x\#y : x \in \Sigma^*, y \in L1, |x| = |y| \}$

CSE331: Automata and Computability

Assignment 2

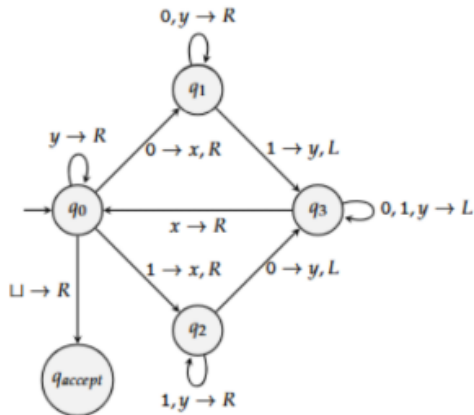
Deadline: 12/08/2026 11:59 PM

Construct a PDA for L2.



Turing Machine

A. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains equal numbers of 0s and 1s}\}$, where $\Sigma = \{0, 1\}$

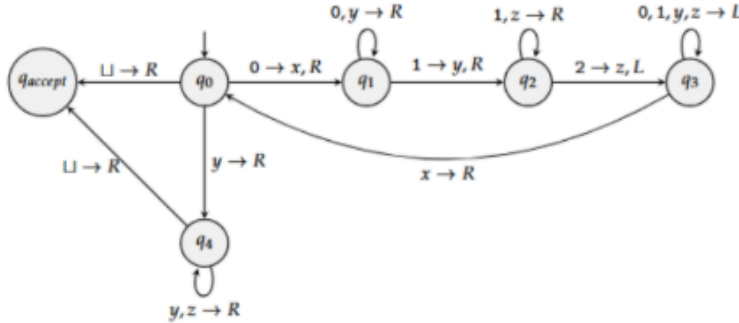


CSE331: Automata and Computability

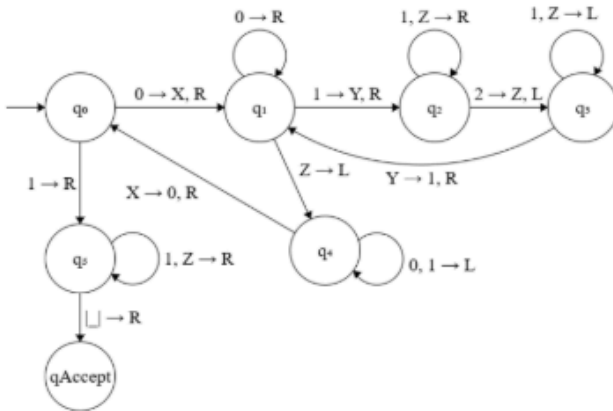
Assignment 2

Deadline: 12/08/2026 11:59 PM

B. $L(M) \rightarrow \{0^n 1^n 2^n \mid n \geq 0\}$, where $\Sigma = \{0, 1, 2\}$

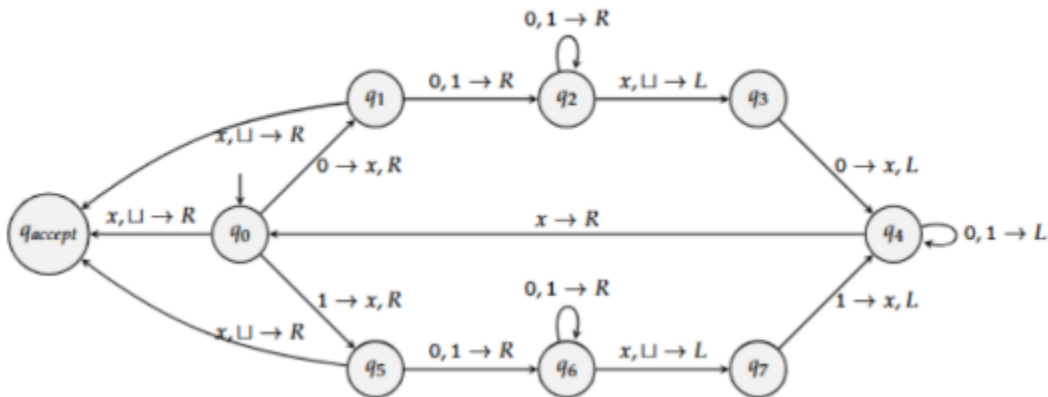


C. $L(M) \rightarrow \{0^i 1^j 2^k \mid i, j, k > 0 \text{ and } k = i * j\}$, where $\Sigma = \{0, 1, 2\}$



D. $L(M) \rightarrow \{0^{2^n} \mid n \geq 0\}$, where $\Sigma = \{0\}$
See example 3.7 from the book.

E. $L(M) \rightarrow \{w \in \Sigma^+ \mid w \text{ is a palindrome}\}$, where $\Sigma = \{0, 1\}$



CSE331: Automata and Computability

Assignment 2

Deadline: 12/08/2026 11:59 PM

Prove that the following languages are decidable:

- A. $L = \{\langle G \rangle \mid G \text{ is a connected undirected graph}\}$
See example 3.23 from the book.
- B. $L = \{\langle A, B \rangle \mid A \text{ and } B \text{ are DFAs and } L(A) = L(B)\}$
See theorem 4.5 from the book.

Show that the following statements are true

- A. The collection of decidable languages is closed under Union.
See problem 3.15.a from the book.
- B. The collection of decidable languages is closed under Complement.
For any decidable languages L , let M be the TM that decides L . We construct a TM M' that decides the complement of L .
 M' = "On input w :
 1. Simulate M on w .
 2. If M rejects, accept. Otherwise, reject."Since M is a decider, M' must be a decider.

Reducibility

Given $A_{TM} \rightarrow \{\langle M, w \rangle \mid M \text{ is a TM that accepts } w\}$ is undecidable, prove that the following languages are undecidable.

- A.
 $RL_{TM} \rightarrow \{\langle M, w \rangle \mid M \text{ is a TM that accepts } w\}$
- B.
 $M_{000} \rightarrow \{\langle M \rangle \mid M \text{ is a TM that accepts } 000\}$

CSE331: Automata and Computability

Assignment 2

Deadline: 12/08/2026 11:59 PM

A. $RL_{TM} \rightarrow \{\langle M, w \rangle \mid M \text{ is a TM such that } L(M) \text{ is a regular language}\}$

See theorem 5.3 from the book.

B. $M_{000} \rightarrow \{\langle M \rangle \mid M \text{ is a TM that accepts } 000\}$

Assume that the TM M decides M_{000} and let A be the TM that decides A_{TM}

$A =$ "On input $\langle M, w \rangle$:

1. Construct the following machine M_1 :

$M_1 =$ "On input x :

1. If $x \neq 000$, reject.

2. If $x = 000$, run M on w and accept if M accepts; otherwise reject."

2. Run M on $\langle M_1 \rangle$.

3. If M accepts, accept; otherwise reject."

As M is a decider, A is also a decider. This forms a contradiction.

Hence, as A_{TM} is undecidable, M_{000} cannot be decided.